DO NOW — RECALL

Packet switching, circuit switching and standards

Name: _____ Class: _____ Date: _____

1. Which TCP/IP layer contains TCP?

2. What does DNS usually provide when given a domain name?

3. Can a graph have more than one possible path between two vertices?

4. Name one advantage of sharing a network link instead of reserving it for one user.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Transport layer.

2. Model answer

An IP address for the name, when the appropriate DNS records exist.

3. Model answer

Yes.

4. Model answer

Capacity can be used by other traffic when a user is idle, improving efficiency.

Vocabulary — know these terms

packet switching — Data is split and forwarded across shared network links.

circuit switching — Communication uses a pre-established reserved path.

sequence information — Control information that allows ordered reconstruction.

retransmission — Sending missing or unacknowledged data again.

standard — Agreed specification supporting compatibility and interoperability.

NEW LEARNING

Read, annotate and explain

Split a message into packets

The sender can split data into separate packets or segments. Each includes control information such as addressing and/or sequence information at relevant protocol layers.

Packets may take different routes

Packet switching lets network traffic share links. IP forwarding can choose different paths; it does not guarantee the same arrival order.

Restore order when needed

TCP provides a reliable ordered byte stream, using acknowledgements and retransmission if required. Do not confuse TCP guarantees with best-effort IP delivery.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Two TCP segments with sequence positions 1 and 2 travel on different routes and arrive 2, then 1. Explain the final data order.

Route

The network may forward them along different paths, so arrival order is not guaranteed.

Receive

TCP uses sequence information to buffer out-of-order data.

Deliver

When missing earlier bytes arrive, TCP makes data available to the application in the correct order: first 1, then 2.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Give one difference between packet switching and circuit switching. [2]

Question 2

Data labelled 1,2,3 arrives as 2,1,3 using TCP. Explain how TCP handles the out-of-order arrival and what the application receives. [4]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain one advantage of packet switching and why communication standards are important for different manufacturers' devices. [4]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

IP guarantees that packets always arrive in the order sent.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.